

PRACTICAL 7: Data Bound controls

AIM:

Create a webpage that demonstrates the use of Data Bound Controls in ASP.NET

SOFTWARE REQUIREMENTS

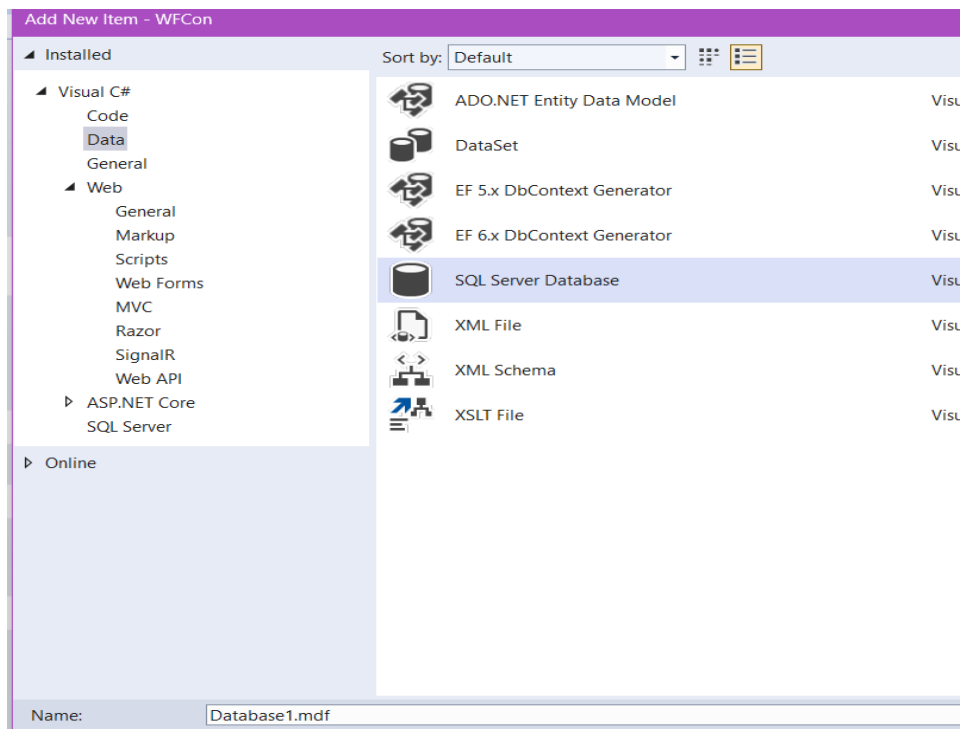
- Visual Studio 2019
- SQL Server / LocalDB
- ASP.NET Web Forms

STEP 1: CREATE PROJECT

1. Open Visual Studio
2. Click **Create New Project**
3. Select:
ASP.NET Web Application (.NET Framework)
4. Click Next
5. Select:
Web Forms
6. Enter Project Name: DataControls
7. Click Create

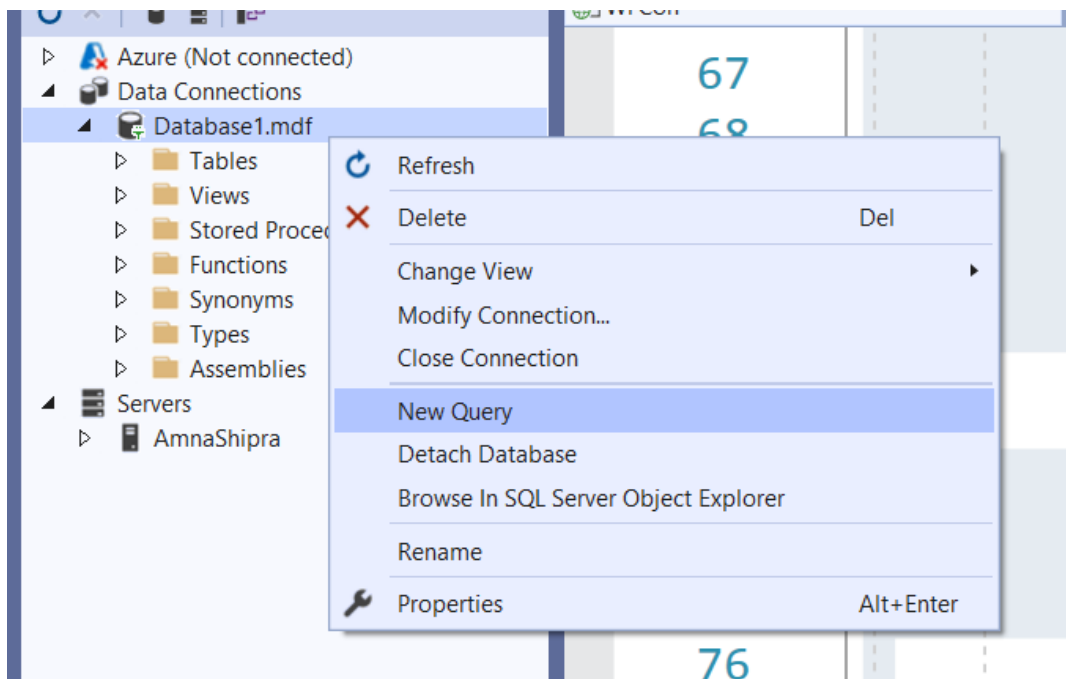
STEP 2: Add Database

1. Right-click Project → **Add** → **New Item**
2. Select **SQL Server Database**
3. Name → Database1.mdf



Step 3: Create Table

Right Click on Database → New Query



Run this query:

```
CREATE TABLE DemoData
(
    Id INT IDENTITY(1,1) PRIMARY KEY,
    Name VARCHAR(100),
```

```
City VARCHAR(100),  
Course VARCHAR(100)  
);
```

Click Update Database

STEP 4: ADD WEB FORM

1. Right click Project → Add → Web Form
2. Name: WebForm1.aspx

STEP 5: Design UI / FRONTEND CODE

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"  
Inherits="DataControls.WebForm1" %>  
  
<!DOCTYPE html>  
  
<html>  
<head runat="server">  
    <title></title>  
</head>  
<body>  
    <form id="form1" runat="server">  
        <div>  
            <asp:Label ID="Label1" runat="server" Text="Name"></asp:Label>  
<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>  
<br /><br />  
  
<asp:Label ID="Label2" runat="server" Text="City"></asp:Label>  
<asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>  
<br /><br />  
  
<asp:Label ID="Label3" runat="server" Text="Course"></asp:Label>  
<asp:TextBox ID="TextBox3" runat="server"></asp:TextBox>  
<br /><br />  
  
<asp:Button ID="Button1" runat="server" Text="Insert Data" OnClick="Button1_Click" />  
&nbsp;   <asp:Button ID="Button2" runat="server" Text="Show Data" OnClick="Button2_Click" />  
  
<br /><br />  
  
<!-- GridView -->  
<asp:GridView ID="GridView1" runat="server"></asp:GridView>  
<br />  
  
<!-- DataList -->  
<asp:DataList ID="DataList1" runat="server" RepeatColumns="2">  
    <ItemTemplate>  
        <%# Eval("Name") %> - <%# Eval("City") %> - <%# Eval("Course") %><br />  
    </ItemTemplate>  
</asp:DataList>  
<br />
```

```

<!-- Repeater -->
<asp:Repeater ID="Repeater1" runat="server">
    <ItemTemplate>
        <%# Eval("Name") %> | <%# Eval("City") %> | <%# Eval("Course") %><br />
    </ItemTemplate>
</asp:Repeater>
<br />

<!-- DetailsView -->
<asp:DetailsView ID="DetailsView1" runat="server"></asp:DetailsView>
<br />

<!-- FormView -->
<asp:FormView ID="FormView1" runat="server">
    <ItemTemplate>
        Name: <%# Eval("Name") %><br />
        City: <%# Eval("City") %><br />
        Course: <%# Eval("Course") %><br />
    </ItemTemplate>
</asp:FormView>
<br />

<!-- ListView -->
<asp:ListView ID="ListView1" runat="server">
    <ItemTemplate>
        <%# Eval("Name") %> - <%# Eval("City") %> - <%# Eval("Course") %><br />
    </ItemTemplate>
</asp:ListView>
    </div>
</form>
</body>
</html>

```

STEP 7: BACKEND CODE

```

using System;
using System.Data;
using System.Data.SqlClient;

namespace DataControls
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        string str = @"Data Source=(LocalDB)\MSSQLLocalDB;
                      AttachDbFilename=|DataDirectory|\Database1.mdf;
                      Integrated Security=True";

        // INSERT DATA
        protected void Button1_Click(object sender, EventArgs e)
        {
            SqlConnection con = new SqlConnection(str);

            string qry = "INSERT INTO DemoData(Name, City, Course) VALUES('" +
                TextBox1.Text + "', '" +
                TextBox2.Text + "', '" +
                TextBox3.Text + "')";

            SqlCommand cmd = new SqlCommand(qry, con);

            con.Open();
            cmd.ExecuteNonQuery();

```

```

        con.Close();

        Response.Write("Data Inserted Successfully");
    }

    // SHOW DATA IN ALL CONTROLS
    protected void Button2_Click(object sender, EventArgs e)
    {
        SqlConnection con = new SqlConnection(str);

        SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM DemoData", con);

        DataSet ds = new DataSet();
        da.Fill(ds);

        // Bind all controls

        GridView1.DataSource = ds;
        GridView1.DataBind();

        DataList1.DataSource = ds;
        DataList1.DataBind();

        Repeater1.DataSource = ds;
        Repeater1.DataBind();

        DetailsView1.DataSource = ds;
        DetailsView1.DataBind();

        FormView1.DataSource = ds;
        FormView1.DataBind();

        ListView1.DataSource = ds;
        ListView1.DataBind();
    }
}

```

STEP 8: RUN APPLICATION

1. Press F5
2. Enter Name, City, Course
3. Click Insert Data
4. Click Show Data

OUTPUT

Name

City

Course

Insert Data

Show Data

Id	Name	City	Course
1	Amna	Mumbai	MCA
2	Abc	Bandra	BCA
3	Pqr	Andheri	BSc CS
4	aaaa	Mumbai	MCA

Amna - Mumbai - MCA Pqr - Andheri - BSc CS

Abc - Bandra - BCA aaaa - Mumbai - MCA

Amna | Mumbai | MCA

Abc | Bandra | BCA

Pqr | Andheri | BSc CS

aaaa | Mumbai | MCA

Id	1
Name	Amna
City	Mumbai
Course	MCA

Name: Amna

City: Mumbai

Course: MCA

Amna - Mumbai - MCA

Abc - Bandra - BCA

Pqr - Andheri - BSc CS

aaaa - Mumbai - MCA

WORKING OF CONTROLS

GridView

Displays data in tabular format with rows and columns

DataList

Displays data in repeated pattern (supports multiple columns)

Repeater

Lightweight control for fully customized display

DetailsView

Displays one record at a time

FormView

Displays one record with custom layout

ListView

Flexible control supporting advanced layouts